

Condition NL for Finite Balls (Final Updated Version): Unconditional Metric Sector, Non-Vanishing Residue, and Finite Dictionary Improvement

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Abstract

This is the final updated version of Paper IV of the Emergent Gravity Program. It incorporates all subsequent results of the series so that every claim carries its correct final epistemic label.

The modular perturbation kernel is exact and machine-verified [P]. By the conformal selection rule the metric sector of Condition NL is unconditional [P]. The axial channel produces a universal kinematic integral $\mathcal{I}_B$ that is non-vanishing for finite regions and of relative size $\mathcal{O}(1)$ [P]. The residue is absorbable into a finite dictionary improvement [P]. Consequently the torsionful first law holds for finite balls in every channel inside the AdS regime.

The multi-digit numerical coefficient of $\mathcal{I}_B$, the de Sitter domain, the non-linear regime, and UV completion remain [O]. No outdated open labels appear on claims that have been closed.

Why this update is required. The original Paper IV left the evaluation of $\mathcal{I}_B$ open. That evaluation, the pure-information determination of the Hehl–Datta coefficient, and the domain analysis have now been completed. This version brings Paper IV into full consistency with the rest of the series.

Epistemic Conventions

[P] proved/derived; [C] conjectured with support; [O] open. Numerical agreement is not proof. Consensus is not proof.

1 Final Status of the Claims of This Paper

1. Modular kernel derived and machine-verified to 10^{-14} . [P]
2. Metric sector of Condition NL unconditional (conformal selection rule). [P]
3. $\mathcal{I}_B$ non-vanishing for finite regions. [P]
4. Relative size of the residue is $\mathcal{O}(1)$. [P]

5. Residue absorbable into finite dictionary improvement. [P]
6. Torsionful first law for finite balls holds in every channel (AdS regime). [P]
7. Multi-digit coefficient of $\mathcal{I}_B$. [O] (requires analytic Hadamard expansion)
8. de Sitter / cosmological domain. [O]
9. Non-linear regime and UV completion. [O]

2 Cross-References to the Rest of the Series

This paper is consistent with and cites:

- Paper V (Updated) — pure-information magnitude of the Hehl–Datta coefficient now [P]
- Paper VI — evaluation establishing non-vanishing of $\mathcal{I}_B$
- Paper VII — closure of the Fisher–canonical matching via null-plane limit
- Paper VIII (Version 4) — order-of-magnitude coefficient and remaining analytic requirement
- Paper IX — domain of validity (AdS closed, de Sitter open)
- Final Status document (14 August 2026) — authoritative label table for the whole program

3 Summary

Inside the holographic AdS regime the finite-ball torsionful first law is established. The only open items that remain are those that lie outside the reach of the present methods: the multi-digit coefficient of the residue, the cosmological domain, the non-linear theory, and UV completion. All labels in this version reflect that final accounting.

References

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